



**RAJALAKSHMI
ENGINEERING COLLEGE**

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING LAB MANUAL**

CS23432 – Software Construction

(REGULATION 2023)

**RAJALAKSHMI ENGINEERING COLLEGE
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Year / Branch / Section: 2nd B.tech AIML D

Semester: IV

Academic Year: 2026

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9.		Load Testing and Pipelines.
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EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

Aim:

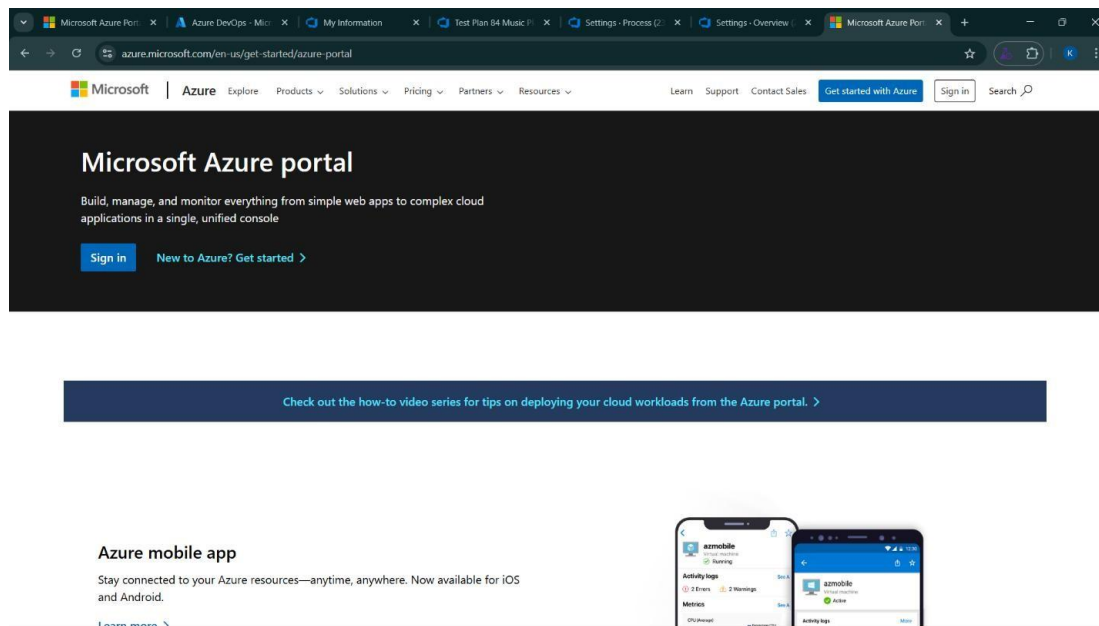
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

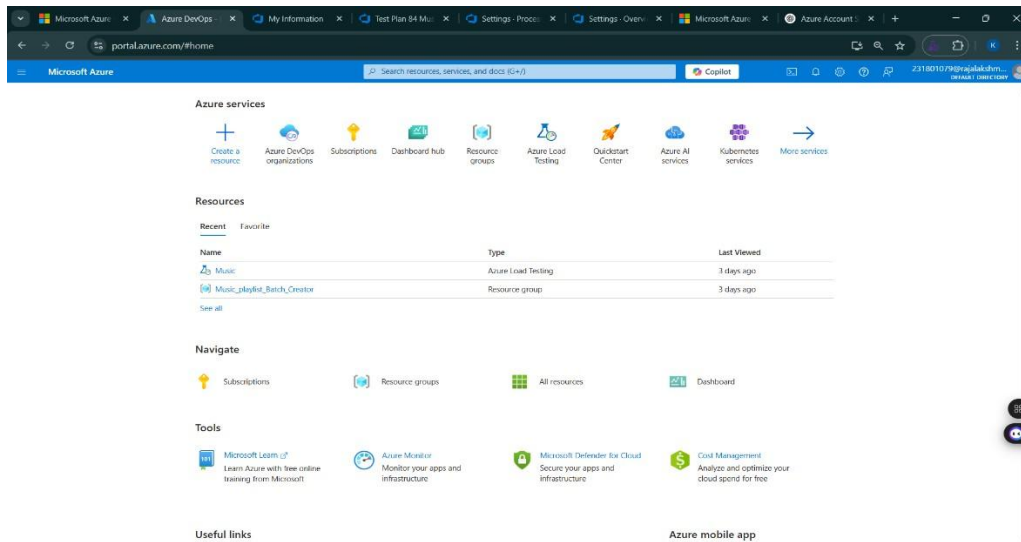
1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/getstarted/azure-portal>.

Sign in using your Microsoft account credentials.

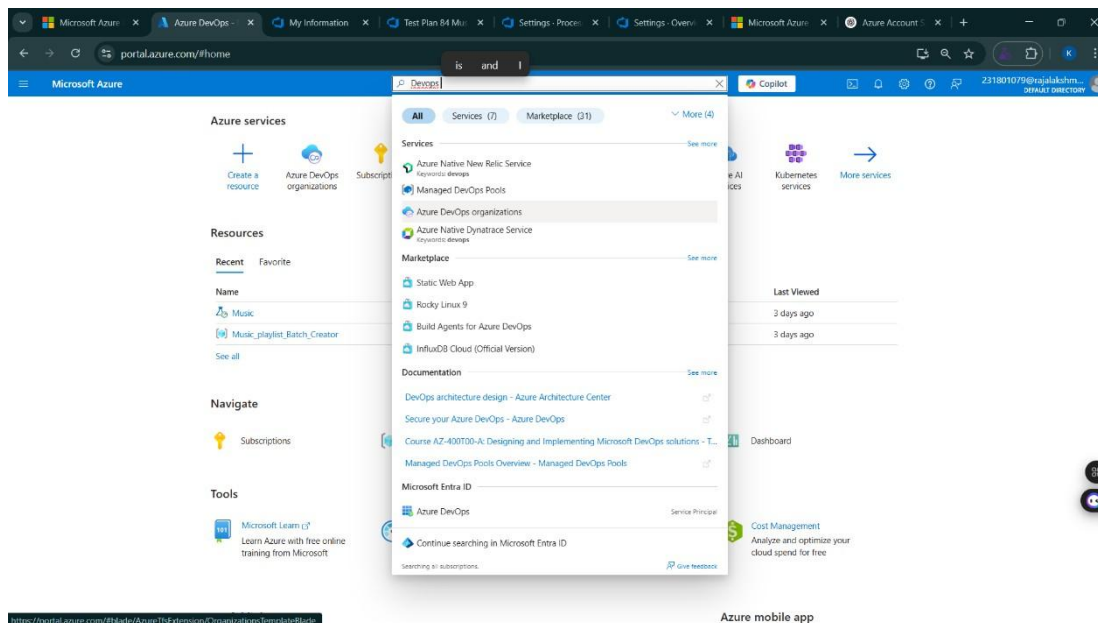
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



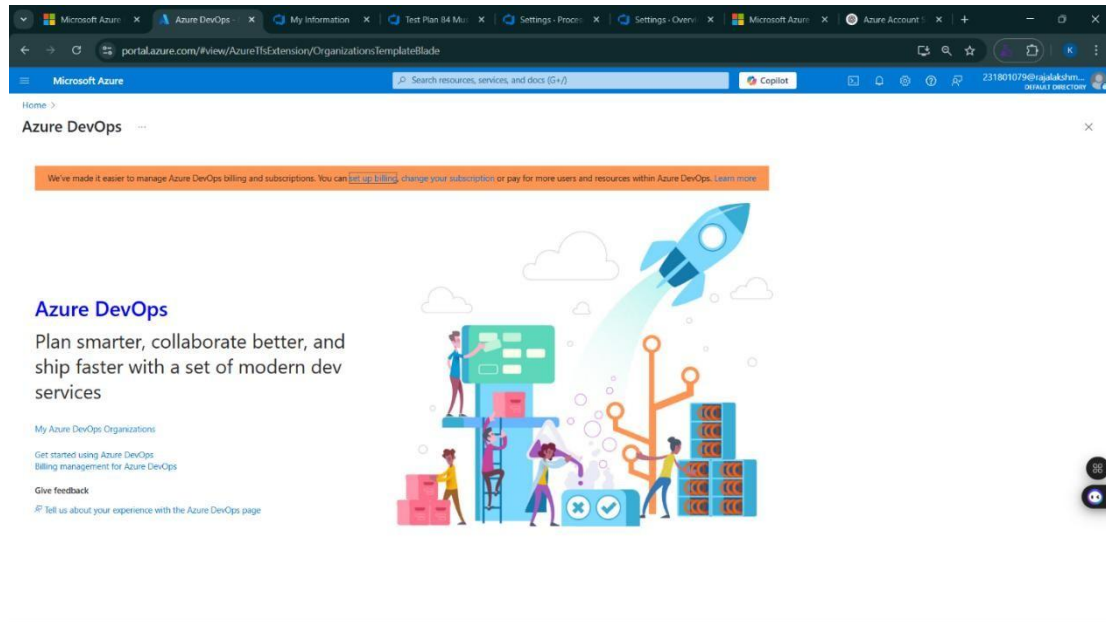
2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the *My Azure DevOps Organization* link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

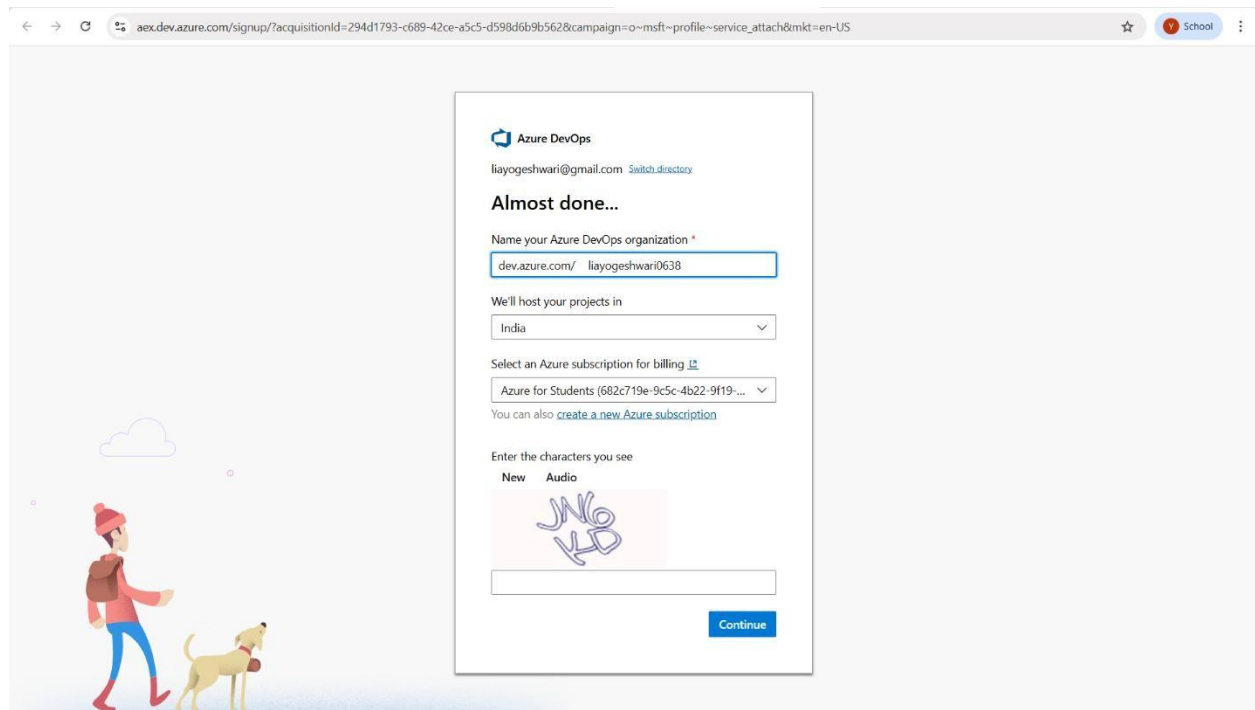
EXP NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STROY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



← → ↻ aex.dev.azure.com/signup/?acquisitionId=294d1793-c689-42ce-a5c5-d598d6b9b562&campaign=o~msft~profile~service_attach&mkt=en-US ☆ School

Azure DevOps
liyogeshwari@gmail.com [Switch directory](#)

Almost done...

Name your Azure DevOps organization *

dev.azure.com/ liyogeshwari0638

We'll host your projects in

India

Select an Azure subscription for billing [link](#)

Azure for Students (682c719e-9c5c-4b22-9f19-...

You can also [create a new Azure subscription](#)

Enter the characters you see

New Audio

JNG KD

Continue

2.Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

Create new project

Project name *

Music Playlist Batch Creator

Description

Visibility



Public

Anyone on the internet can view the project. Certain features like TFVC are not supported.



Private

Only people you give access to will be able to view this project.

By creating this project, you agree to the Azure DevOps [code of conduct](#)

^ Advanced

Version control ?

Git

Work item process ?

231801095 Agile

Cancel

Create

3.Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

STUDENT REG NO

CS23432

← → ↻ aex.dev.azure.com/me?mkt=en-US ☆ Y Scho

Microsoft Yogeshwari .P Sign out



Yogeshwari .P Edit profile
liayogeshwari@gmail.com

Microsoft account

India
liayogeshwari@gmail.com

Azure DevOps Organizations

Create new organization

dev.azure.com/liayogeshwari (Owner)

Projects	Actions
 Food control	Open in Visual Studio
New project	

Visual Studio Dev Essentials

Get everything you need to build and deploy your app on any platform.

[Use your benefits](#)

4. Project dashboard

← → ↻ dev.azure.com/liayogeshwari/Food%20control/_dashboards/dashboard/f178bdb9-c4e2-4602-afc6-ca385e6678c0 ☆ Y Scho

Azure DevOps liayogeshwari / Food control / Overview / Dashboards

Food control + Food control Team - Overview Modified Just now by Yogeshwari .P Edit



This dashboard doesn't have widgets just yet!

Add one or more widgets to gain visibility into your team's progress.

[Add a widget](#)

Don't miss out
[https://dev.azure.com/liayogeshwari/Food control](https://dev.azure.com/liayogeshwari/Food%20control)

5.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

dev.azure.com/liayogeshwari/Food%20control/_backlogs/backlog/Food%20control%20Team/Issues

Azure DevOps liayogeshwari / Food control / Boards / Backlogs

Food control Team

+ New Work Item View as Board Column Options

Backlog Analytics

Order	ID	Title	Assigned To	State	Tags
1	1	> to do list		To Do	

Issues

Planning

Drag and drop work items to include them in a sprint.

Food control Team backlog

Sprint 1
Planned Effort: 3
1 1

+ New Sprint

dev.azure.com/liayogeshwari/Food%20control/_workitems/recentlyupdated/

Azure DevOps liayogeshwari / Food control / Boards / Work items

Work items

Recently updated + New Work Item Open in Queries Column Options Import Work Items Recycle Bin

Filter by keyword Types Assigned to

ID	Title	Assigned To	State
1	to do list	Unassigned	To Do

Microsoft Sign out

Yogeshwari.P
liayogeshwari@gmail.com
My Microsoft account
Switch directory

Sign in with a different account

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for a project named 'Food control'. The left sidebar contains navigation options: Overview, Boards, Work items (selected), Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The 'Work items' section is active, displaying a list of work items. The table has columns for ID, Title, Assigned To, State, Area Path, and Tags. The work items are sorted by 'Recently updated'.

ID	Title	Assigned To	State	Area Path	Tags
5	to do list	Unassigned	New	Food control	
4	location access	Unassigned	New	Food control	
3	as a user, i want no food wastage system as i can save resources	Unassigned	New	Food control	
2	to do list	Unassigned	Done	Food control	
1	to do list	Unassigned	To Do	Food control	

1.Fill in Epics

The screenshot shows the Azure DevOps web interface for a project named 'Food control'. The left sidebar contains navigation links: Overview, Boards, Work items (selected), Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main content area displays a 'Recently updated' list with one item, 'EPIC 5 to do list'. The work item details are shown for 'EPIC 5 to do list', which is a 'New' item in the 'Food control' area, specifically under 'Food control/Sprint 1'. The 'Description' field is empty with a placeholder 'Click to add Description.'. The 'Discussion' section has a comment box with a placeholder 'Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.' and a 'switch to Markdown editor' link. The 'Planning' section includes fields for Priority (2), Risk, Effort, Business Value, and Time Criticality. The 'Deployment' section has a 'Add link' button and a 'Related Work' section with an 'Add link' button. The 'Classification' section shows 'Value area' as 'Business'.

2.Fill in Features

The screenshot shows the Azure DevOps web interface for a project named 'Food control'. The left sidebar contains navigation links: Overview, Boards, Work items (selected), Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main content area displays a 'Recently updated' list with one item, 'FEATURE 4 location access'. The work item details are shown for 'FEATURE 4 location access', which is a 'New' item in the 'Food control' area, specifically under 'Food control/Sprint 1'. The 'Description' field is empty with a placeholder 'Click to add Description.'. The 'Discussion' section has a comment box with a placeholder 'Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.' and a 'switch to Markdown editor' link. The 'Planning' section includes fields for Priority (2), Risk, Effort, Business Value, and Time Criticality. The 'Deployment' section has a 'Add link' button and a 'Related Work' section with an 'Add link' button. The 'Classification' section shows 'Value area' as 'Business'.

3.Fill in User Story Details

The screenshot displays the Azure DevOps web interface for editing a User Story. The browser address bar shows the URL: `dev.azure.com/liayogeshwari/Food%20control/_workitems/edit/3/`. The left sidebar contains a navigation menu with options: Overview, Boards, Work items (selected), Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The main content area is titled 'Recently updated' and shows a 'Back to Work Items' link. The User Story is titled 'USER STORY 3' and has the text '3 as a user, i want no food wastage system as i can save resources'. It is currently in the 'New' state and is associated with the 'Food control' area and 'Food control\Sprint 1' iteration. The story has 0 comments and 0 tags. The right sidebar contains several sections: 'Save', 'Follow', 'Details', 'Deployment' (with a link to 'To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting'), 'Development' (with a link to 'Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.'), and 'Related Work' (with a link to 'Add an existing work item as a parent').

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4

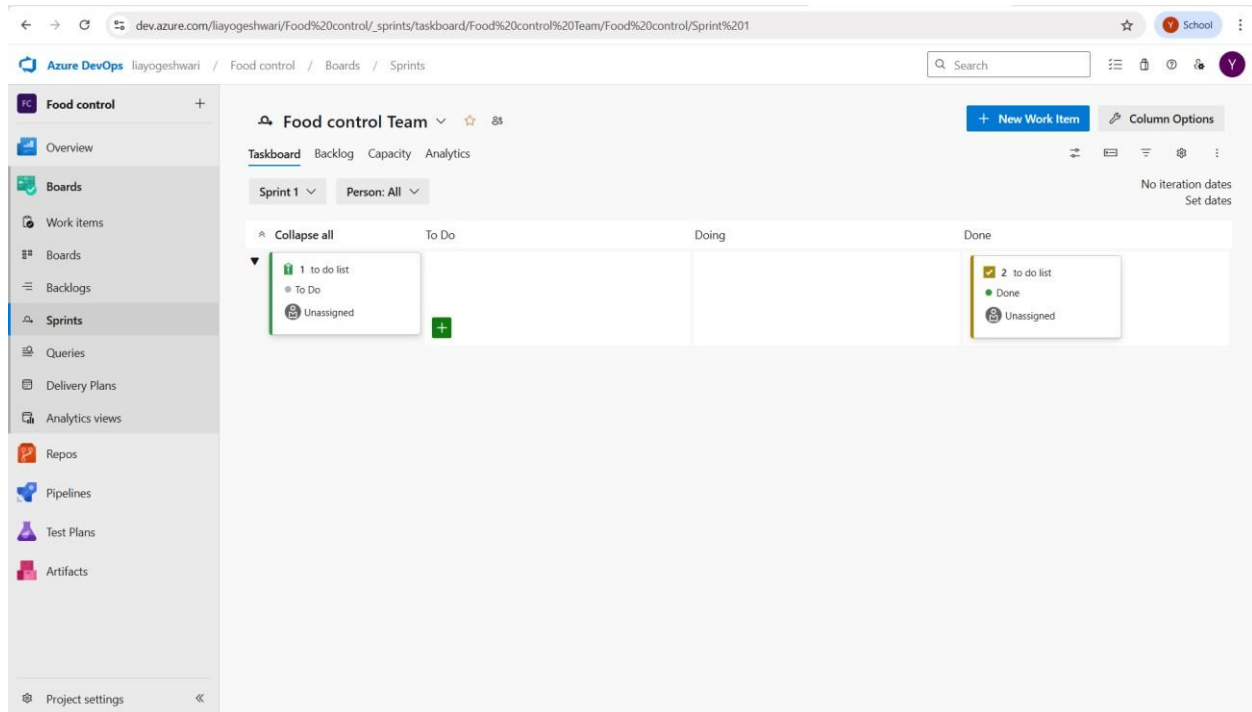
SPRINT PLANNING

Aim:

To assign user story to specific sprint for the Music Playlist Batch Creator Project.

Sprint Planning

Sprint 1



Result:

The Sprints are created for the Music Playlist Batch Creator Project.

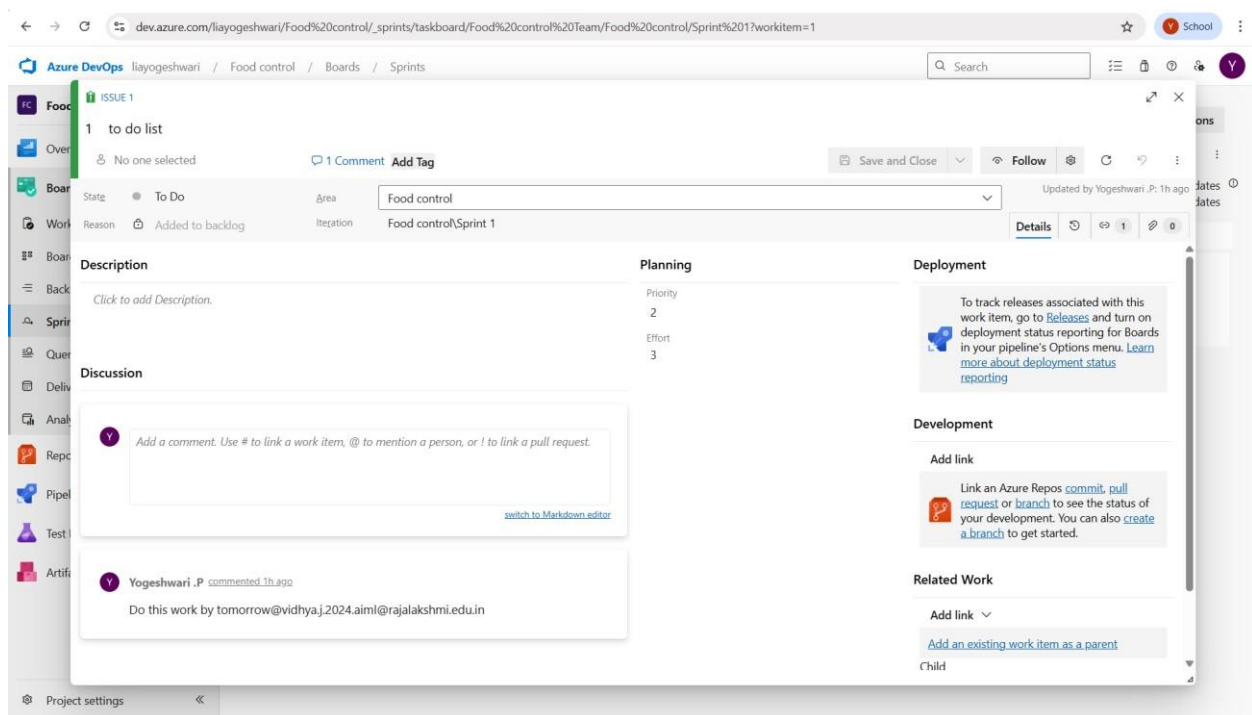
EXP NO: 5

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories - Music Playlist Batch Creator Project.

Poker Estimation



Result:

The Estimation/Story Points is created for the project using Poker Estimation.

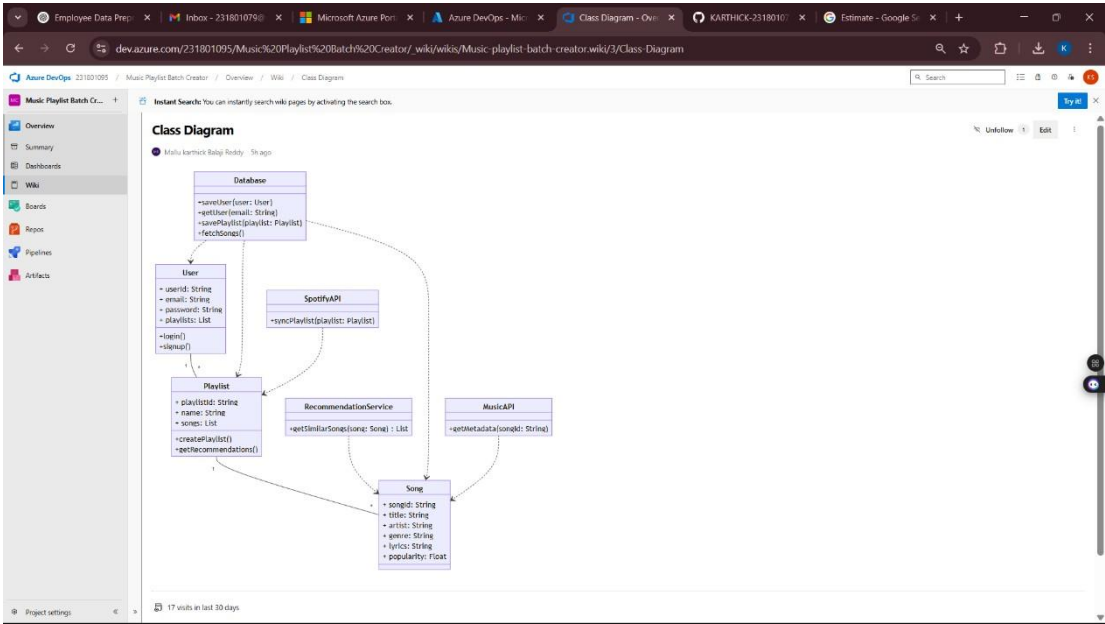
EXP NO: 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

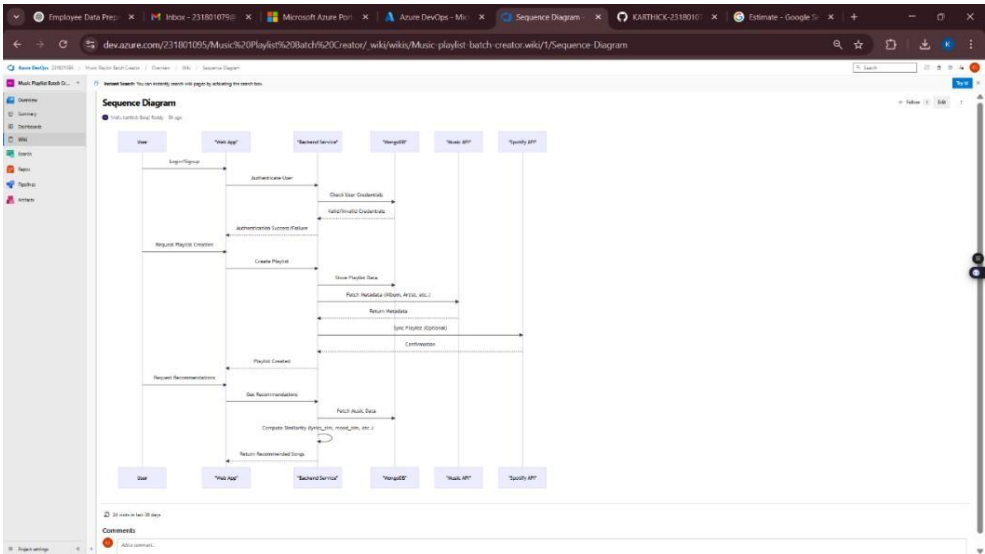
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project. **6A.**

Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the Music Playlist Batch Creator.

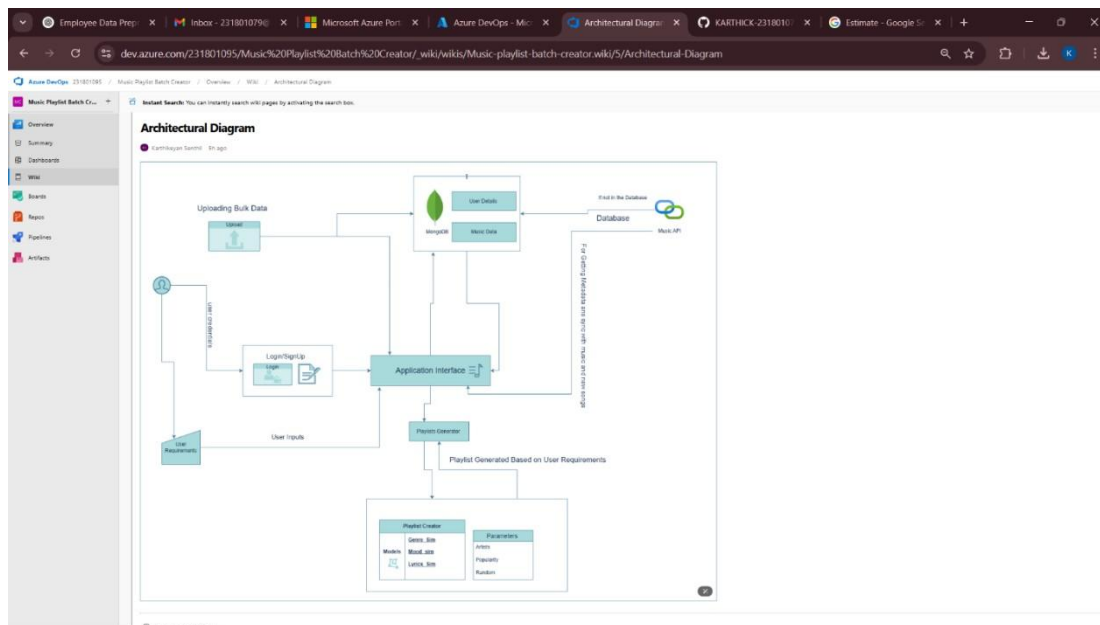
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

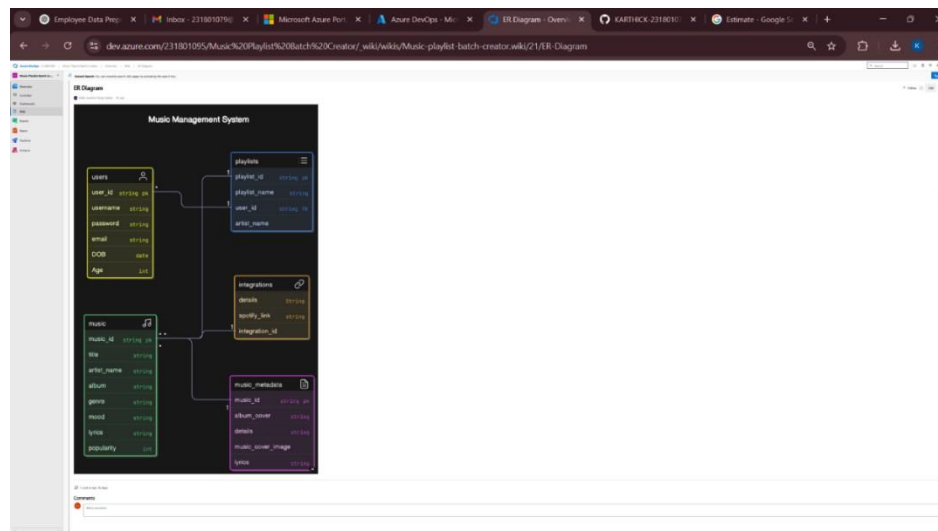
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project. 7A.

Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Music Playlist Batch
Creator

EXP NO: 8

TESTING – TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

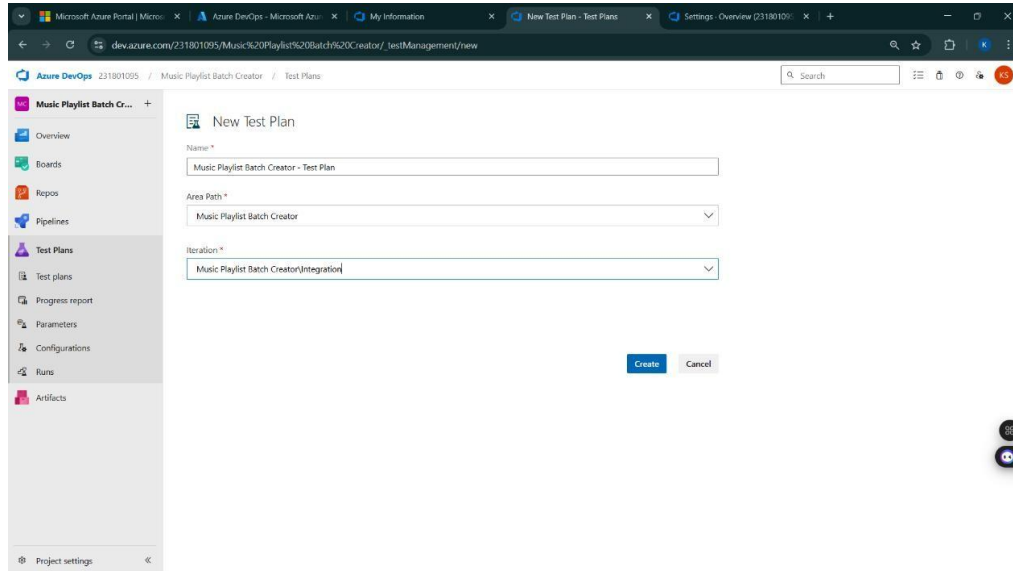
Test Planning and Test Case

Test Case Design Procedure

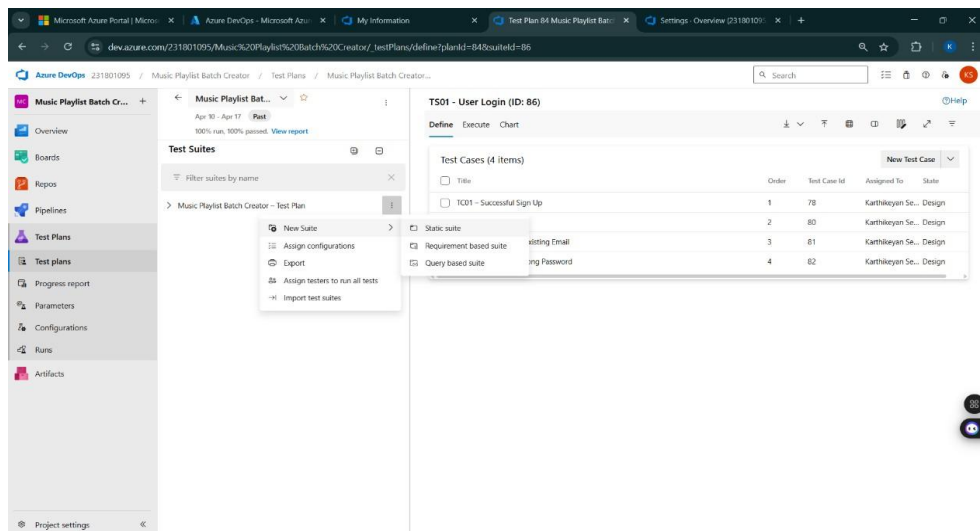
- 1. Understand Core Features of the Application** ○ User Signup & Login ○ Viewing and Managing Playlists ○ Fetching Real-time Metadata ○ Editing playlists (rename, reorder, record)
 - Creating smart audio playlists based on categories (mood, genre, artist, etc.)
- 2. Define User Interactions** ○ Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).
- 3. Design Happy Path Test Cases** ○ Focused on validating that all features function as expected under normal conditions.
 - Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.
- 4. Design Error Path Test Cases** ○ Simulate negative or unexpected scenarios to test robustness and error handling.
 - Example: Login fails with invalid credentials, save fails when offline, no recommendations found.
- 5. Break Down Steps and Expected Results** ○ Each test case contains step-by-step actions and a corresponding expected outcome.
 - Ensures clarity for both testers and automation scripts.
- 6. Use Clear Naming and IDs** ○ Test cases are named clearly (e.g., TC01 – Successful Login, TC10 – Save Playlist Fails).
 - Helps in quick identification and linking to user stories or features.
- 7. Separate Test Suites** ○ Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).
 - Improves organization and test execution flow in Azure DevOps.

8. **Prioritize and Review**
- Critical user actions are marked high-priority.
 - Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test suite



3.Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Music Playlist Batch Creator – Test Plans

USER STORIES

- As a user, I want to sign up and log in securely so that I can access my playlists (ID: 79).

- As a user, I need to see my playlist in one place (ID: 76).
- As a user, I should be able to create an audio playlist as needed (ID: 73).
- As a user, I should be able to rename, record, and change the playlist (ID: 68).
- As a user, I need to have real-time metadata (ID: 65).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. TC01 – Successful Sign Up ○ Action:

- Go to the Sign-Up page.
- Enter valid name, email, and password.
- Click "Sign Up".

○ Expected Results:

- Sign-Up form is displayed.
- Fields accept values without error.
- Account is created, and the user is redirected to the dashboard.

○ Type: Happy Path

2. TC02 – Secure Login

○ Action:

- Go to the Login page.
- Enter valid email and password.
- Click on "Login".

○ Expected Results:

- Login form is displayed.
- Fields accept data without error.
- User is logged in and redirected to the dashboard.

○ Type: Happy Path

3. TC03 – Sign Up with Existing Email

○ Action:

- Go to the Sign-Up page.
- Enter a name and an already registered email.
- Click on "Sign Up".

○ Expected Results:

- Fields accept data.
- Error message "Email already registered" is displayed.

○ Type: Error Path

4. TC04 – Login with Wrong Password

○ Action:

- Go to the Login page.

- Enter valid email and incorrect password.
- Click on "Login".
- **Expected Results:**
 - Input is accepted.
 - Error message "Invalid username or password" is shown.
- **Type:** Error Path

Test Suit: TS02 - View Playlists (ID: 87)

1. TC05 – View Playlist Page

- **Action:**
 - Log in successfully.
 - Navigate to "My Playlists" section.
- **Expected Results:**
 - All created playlists are displayed clearly.
- **Type:** Happy Path

2. TC06 – Playlist Loading Failure

- **Action:**
 - Disconnect from the internet.
 - Navigate to "My Playlists".
- **Expected Results:**
 - Network is offline.
 - Error message "Unable to load playlists" is shown.
- **Type:** Error Path

Test Suit: TS03 - Real-Time Metadata (ID: 88)

1. TC07 – Real-Time Metadata Display ○

Action:

- Play a song.
- Observe the metadata panel.
- **Expected Results:**
 - Metadata (title, artist, album, duration) is displayed and updates in real time.
- **Type:** Happy Path

2. TC08 – Metadata Not Updating ○

Action:

- Play a different song.
- Observe the metadata panel.
- **Expected Results:**
 - Metadata remains static or shows default/fallback message.
- **Type:** Error Path

Test Suit: TS04 - Playlist Editing (ID: 89)

1. TC09 – Rename Playlist Successfully ○

Action:

- Navigate to "My Playlists".
- Click "Rename" next to a playlist.
- Enter a new name and click "Save".

○ **Expected Results:**

- Playlist name updates successfully.

○ **Type:** Happy Path

2. TC10 – Rename with Blank Name ○

Action:

- Click "Rename" on a playlist.
- Leave the field blank.
- Click "Save".

○ **Expected Results:**

- Error message "Playlist name cannot be empty" is shown.

○ **Type:** Error Path

3. TC11 – Change Playlist Order ○

Action:

- Open a playlist.
- Drag and drop songs to reorder.
- Click "Save".

○ **Expected Results:**

- Playlist order is updated and saved.

○ **Type:** Happy Path

4. TC12 – Change Playlist Order Fails ○

Action:

- Login and go to "My Playlists".
- Select a playlist.
- Go offline or simulate server error.
- Reorder songs and click "Save Order".

○ **Expected Results:**

- Error message: "Failed to update order. Please check your connection".

○ **Type:** Error Path

Test Suit: TS05 - Smart Playlist Creation (ID: 90)

1. TC13 – Generate Playlist Based on Various Categories ○

Action:

- Login with valid credentials.

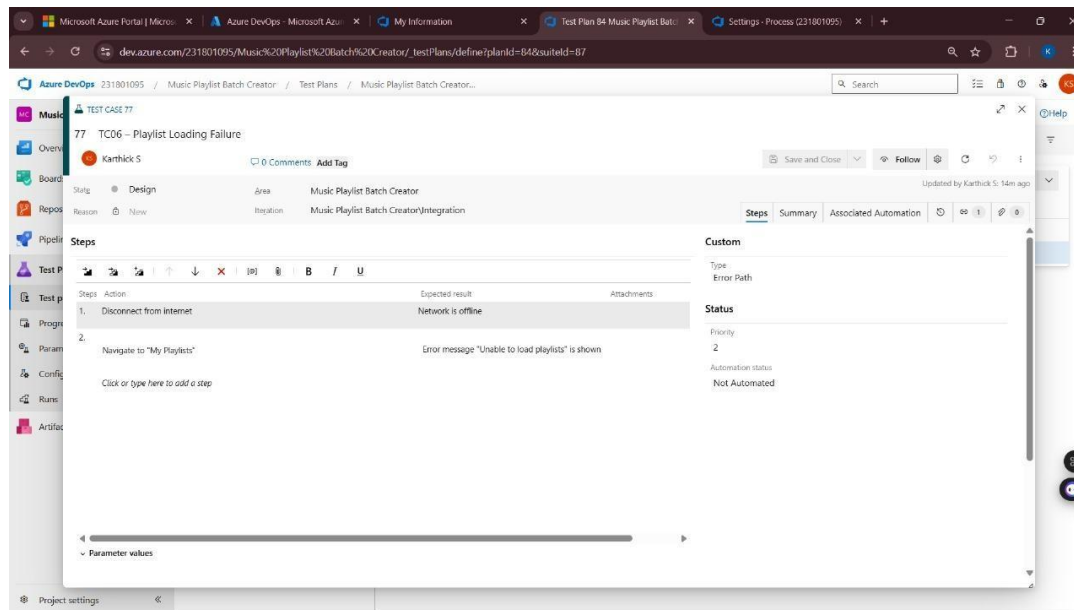
- Click on "Generate Playlist".
- Select categories.
- Click "Generate Playlist".
- **Expected Results:**
 - Playlist is generated based on selected mood and categories.
- **Type:** Happy Path

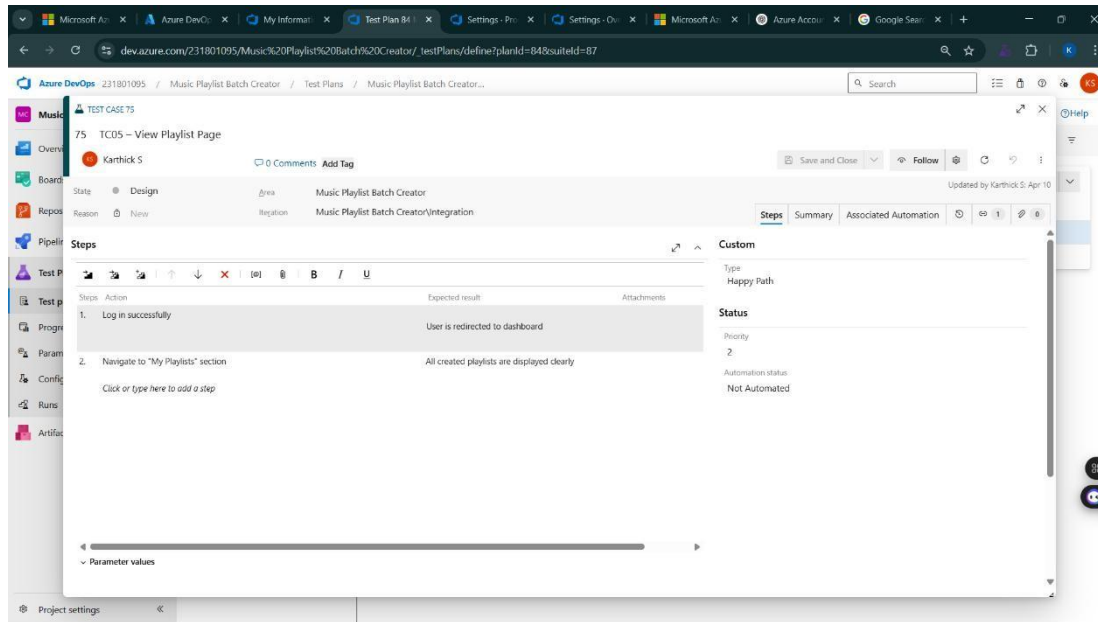
2. TC14 – Fail to Generate Playlist Due to Missing Category Selection or Invalid Input ○

Action:

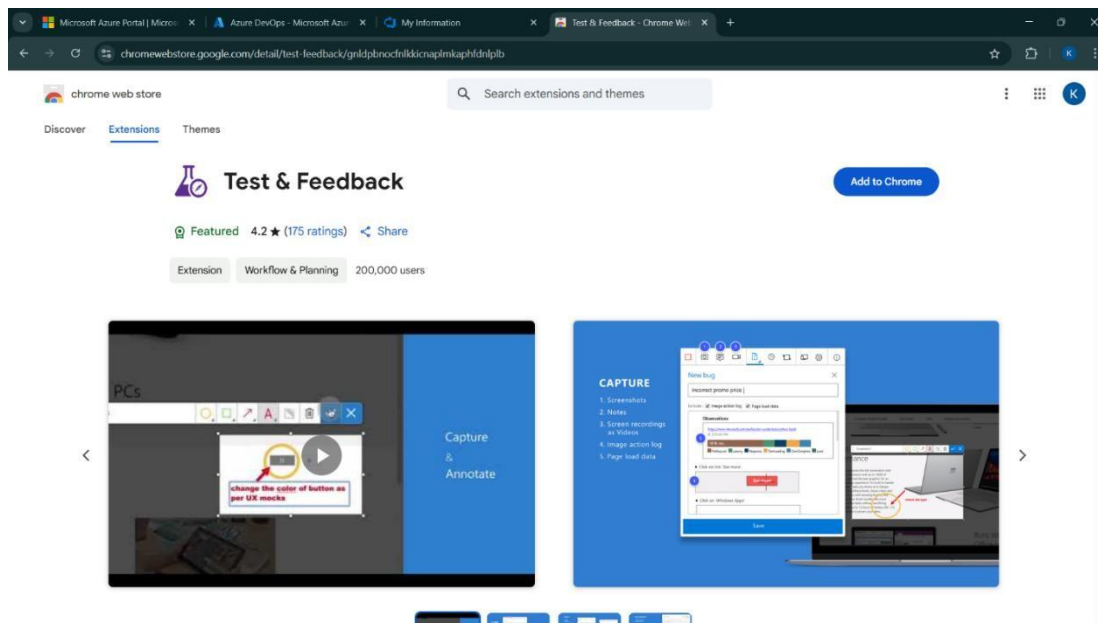
- Login with valid credentials.
- Click on "Generate Playlist".
- Select categories.
- Click "Generate Playlist".
- **Expected Results:**
 - Error message: "Please select at least one valid category" or "No recommendations found for the selected filters".
- **Type:** Error Path

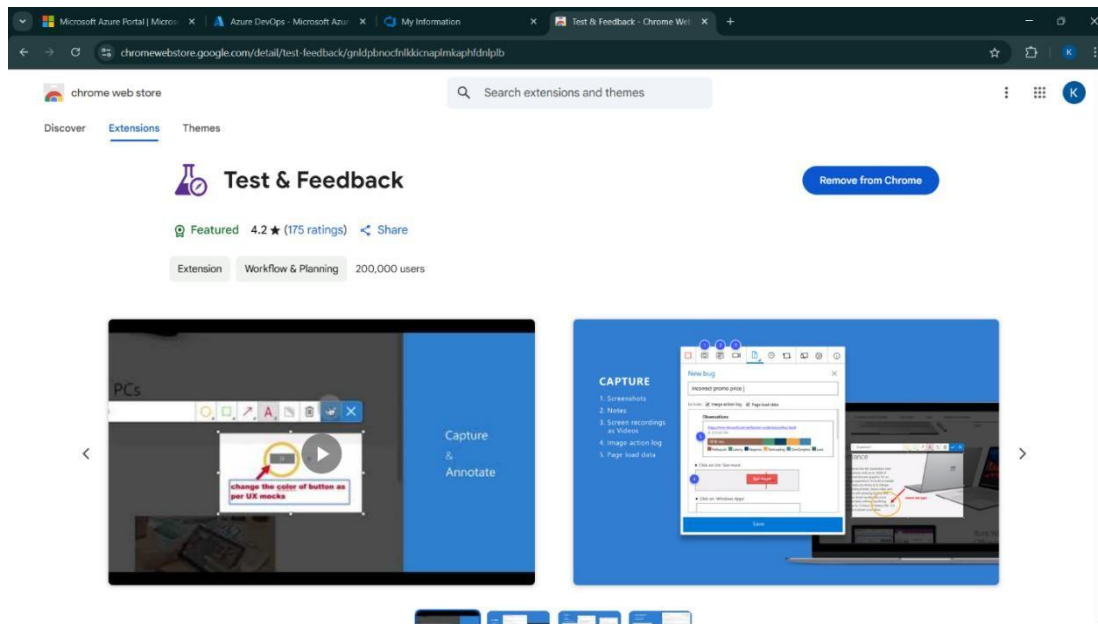
Test Cases



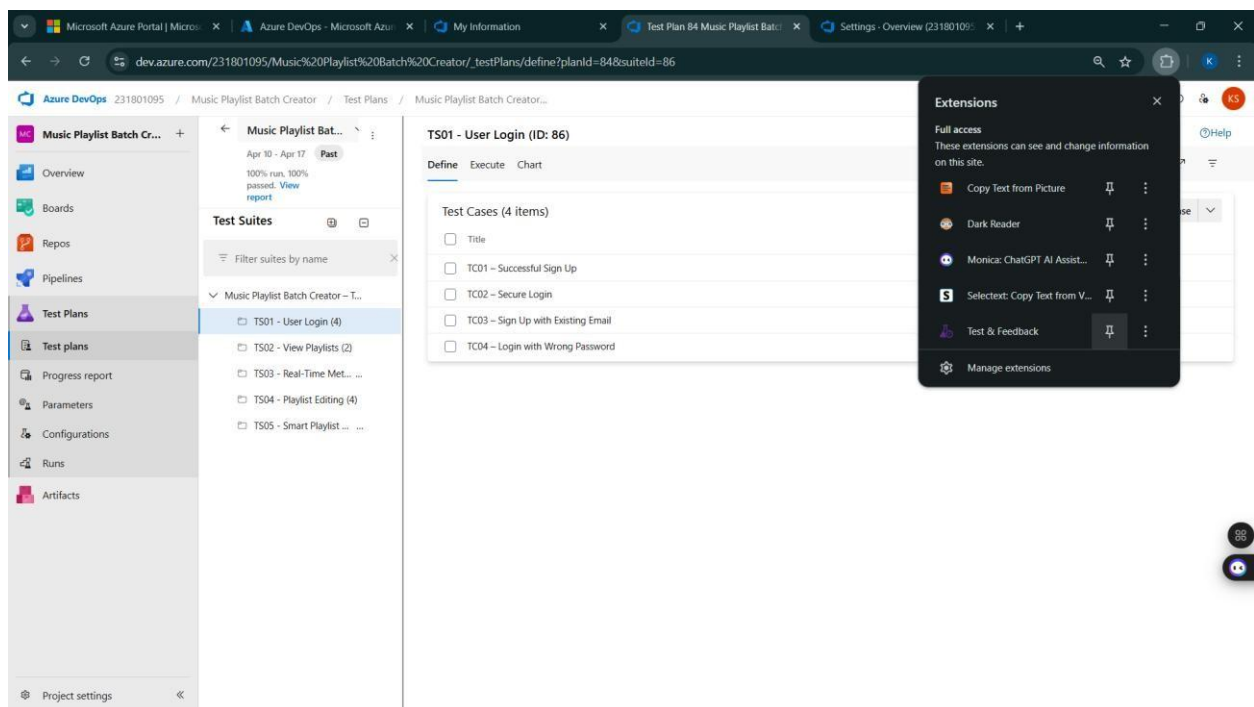


4.Installation of test





Test and feedback
Showing it as an extension



5. Running the test cases

The screenshot shows the Azure DevOps Test Plans interface. The left sidebar contains navigation options: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, and Artifacts. The main area displays the 'Music Playlist Batch Creator' test suite, which is 100% passed. A table titled 'Test Points (2 items)' shows the following data:

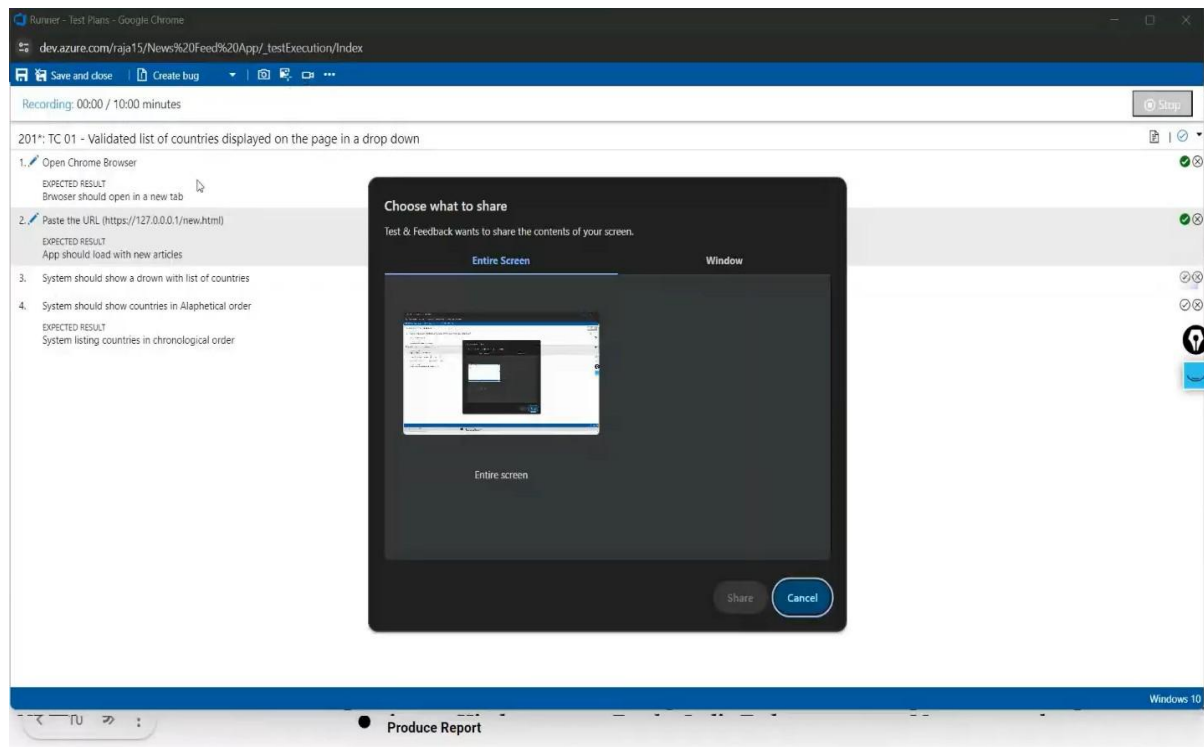
Title	Outcome	Order	Test Case Id	Configuration	Tester
TC05 - View Playlist Page	Passed	1	75	Windows 10	Mallu karthick B...
TC06 - Playlist Loading Failure	Passed	2	77	Windows 10	Mallu karthick B...

A context menu is open for the first test point, showing options: View execution history, Mark Outcome, Run, Reset test to active, Edit test case, Assign tester, and View test result. The 'Run' option is selected, and a sub-menu shows 'Run for web application', 'Run for desktop application', and 'Run with options'.

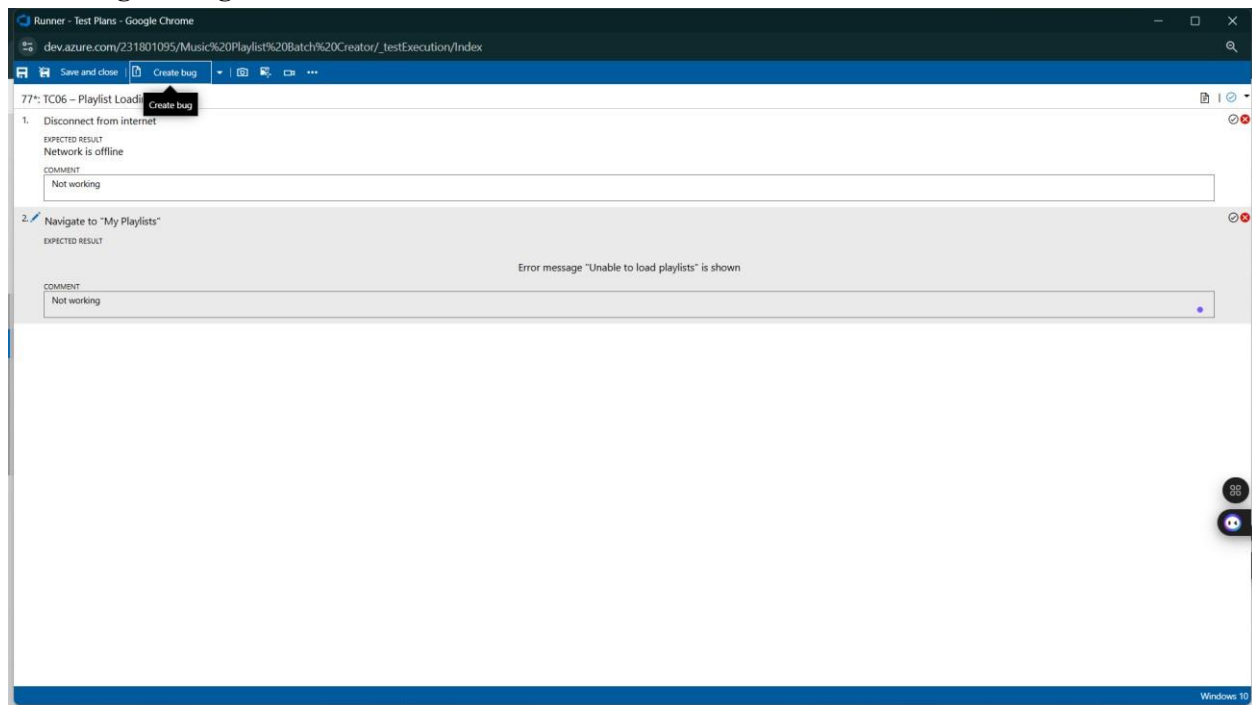
The screenshot shows the 'Runner - Test Plans - Google Chrome' window displaying the test execution results for '75*: TC05 - View Playlist Page'. The test steps and their results are as follows:

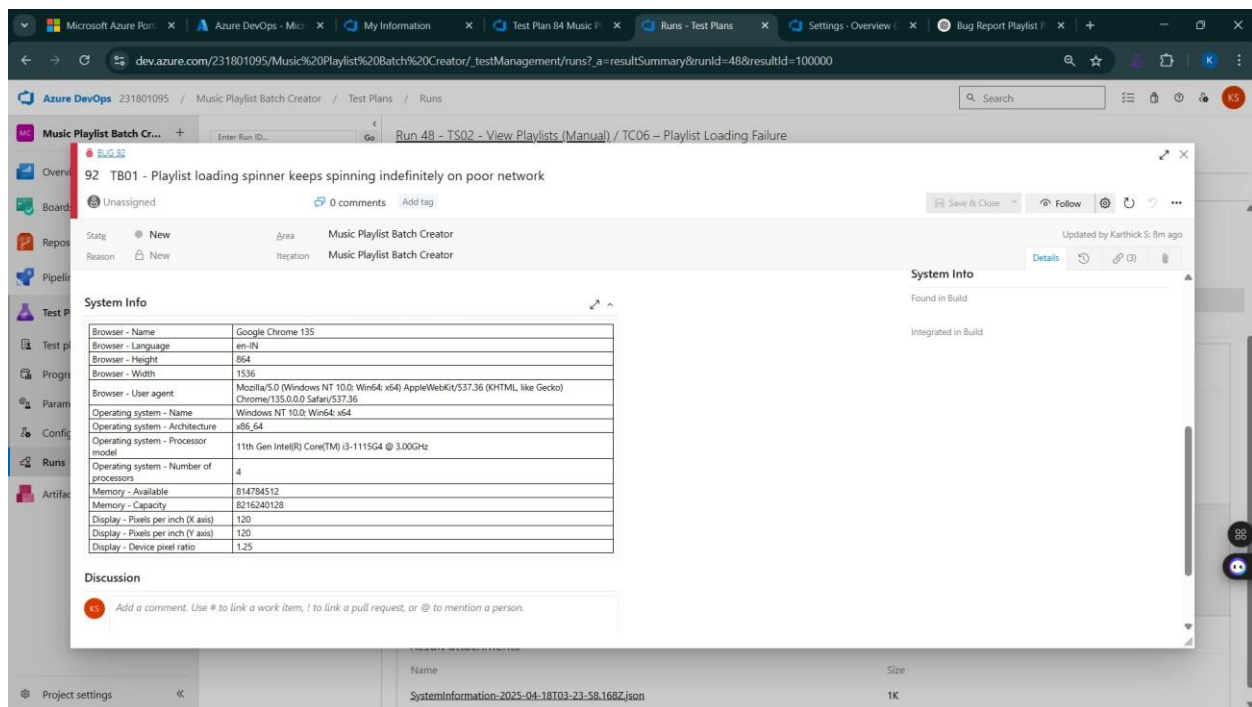
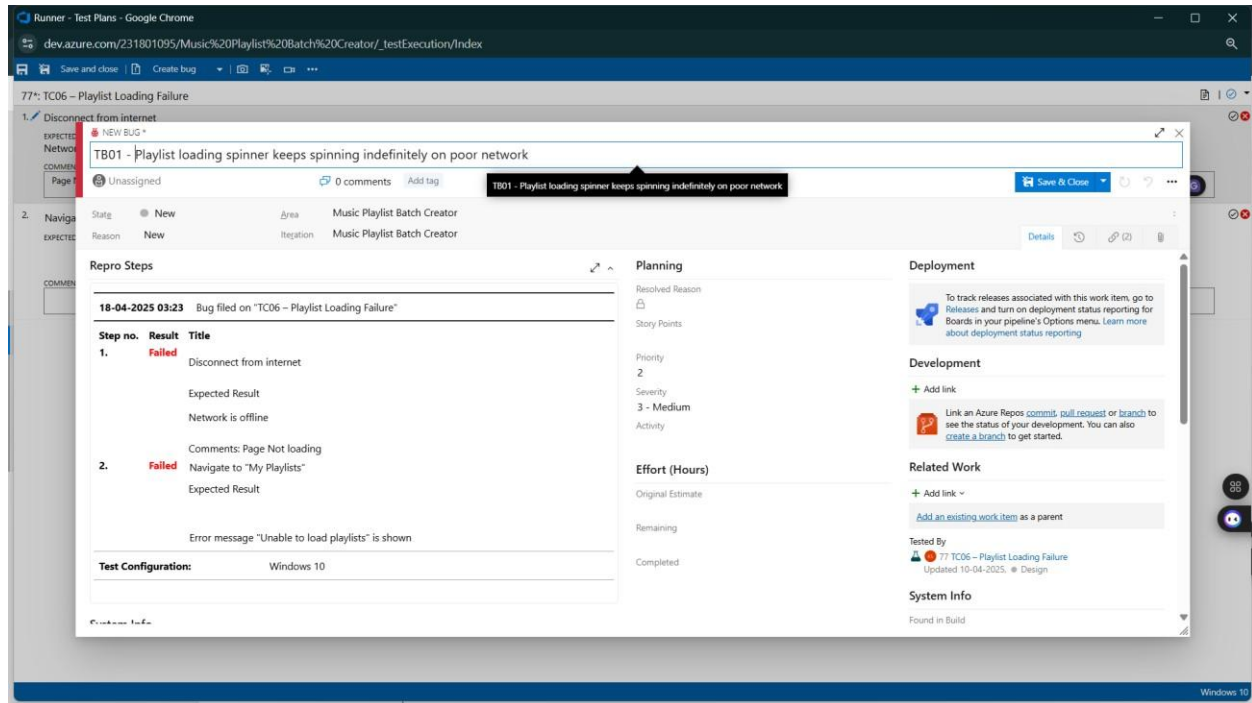
Step	Expected Result	Actual Result
1. Log in successfully	User is redirected to dashboard	Passed
2. Navigate to "My Playlists" section	All created playlists are displayed clearly	Passed

6. Recording the test case



7. Creating the bug





8. Test case results

The screenshot shows the Azure DevOps interface for a project named 'Music Playlist Batch Creator'. The left sidebar contains navigation links for Overview, Boards, Repos, Pipelines, Test Plans, Progress report, Parameters, Configurations, Runs, and Artifacts. The 'Test Plans' section is selected, showing a list of test suites. The 'Test Suites' panel displays a filter by name and a list of suites: TS01 - User Login (4), TS02 - View Playlists (2), TS03 - Real-Time Met..., TS04 - Playlist Editing (4), and TS05 - Smart Playlist ... The 'Test Plans' panel shows 'TS02 - View Playlists (ID: 87)' with two test points: 'TC05 - View Playlist Page' and 'TC06 - Playlist Loading Failure'. The 'Test Case Results' modal is open, showing a table of test results.

Outcome	TimeStamp	Configuration	Run by	Tester	Test Plan
Passed	4m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Not Applicable	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	14m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	Tuesday	Windows 10	Karthikeyan Senthil	Mallu karthick Balaji ...	Music
Passed	Saturday	Windows 10	Mallu karthick Balaji ...	Mallu karthick Balaji ...	Music
Failed	Saturday	Windows 10	Mallu karthick Balaji ...	Mallu karthick Balaji ...	Music
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ...	Music

9. Test report summary

The screenshot shows the Azure DevOps interface for a project named 'News Feed App'. The left sidebar contains navigation links for Overview, Boards, Work items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, Pipelines, Test Plans, and Artifacts. The 'Work items' section is selected, showing a list of work items. The 'Bug 203' work item is selected, showing details for 'BG 01 - Countries Drop down Not Available on the page'. The 'State' is set to 'New' and the 'Reason' is 'New'. The 'Repro Step' section shows three steps: 1. Open Chrome Browser (Passed), 2. Paste the URL (https://127.0.0.1/new.html) (Passed), and 3. App should load with new articles (Failed). The 'Planning' section shows 'Resolved Reason' and 'Story Points'. The 'Deployment' section shows a link to 'Add link' and a note about tracking releases. The 'Development' section shows a link to 'Add link' and a note about linking an Azure Repos commit, pull request, or branch to see the status of your development.

- Assigning bug to the developer and changing state

Microsoft Azure Portal | Microsoft Azure | My Information | Test Plan 84 Music P | Runs - Test Plans | Settings - Overview |

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testManagement/runs?_a=resultSummary&runId=48&resultId=100000

Azure DevOps 231801095 / Music Playlist Batch Creator / Test Plans / Runs

92 TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Karthick S 0 comments Add tag

Save & Close Follow Details

Updated by Karthick S 33m ago

Repro Steps

18-04-2025 03:23 Bug filed on "TC06 - Playlist Loading Failure"

Step no.	Result	Title
1.	Failed	Disconnect from internet
		Expected Result
		Network is offline
2.	Failed	Comments: Page Not loading
		Navigate to "My Playlists"
		Expected Result

Error message "Unable to load playlists" is shown

Test Configuration: Windows 10

Planning

Resolved Reason

Story Points

Priority 2

Severity 3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

+ Add link

Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Related Work

+ Add link

Add an existing work item as a parent

Tested By

77 TC06 - Playlist Loading Failure

Updated 10-04-2025. @ Design

System Info

Comment Page Not loading

2. Navigate to "My Playlists"

10. Progress report

Microsoft Azure Portal | Microsoft Azure | My Information | Progress report - Test Plans | KARTHIK- 231801079/Music P |

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testManagement/analytics/progressreport

Azure DevOps 231801095 / Music Playlist Batch Creator / Test Plans / Progress report

Music Playlist Batch Creator - Test Plan

Summary

1 Test plans

14 Test points

14 (14 / 14) Test points run

100% Run

100% (14 / 14) Pass rate

14 Passed

Outcome trend

Last 14 Days

Test points

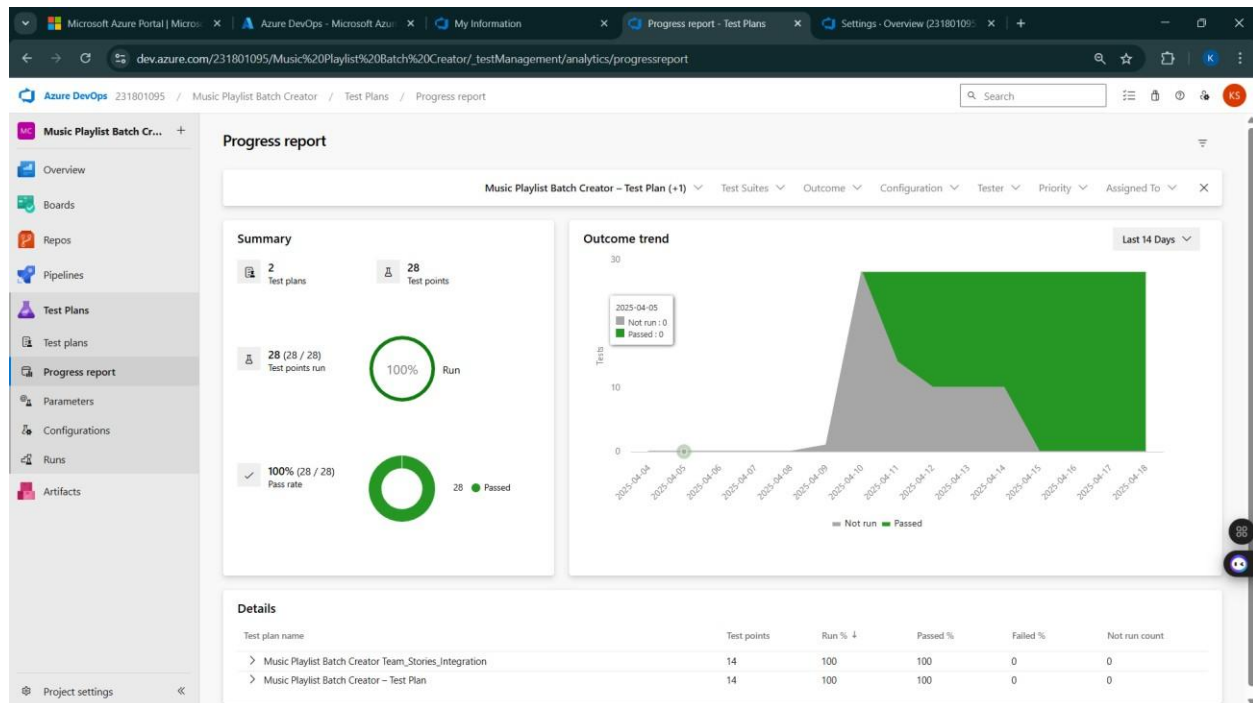
Run %

Passed %

Failed %

Not run count

Test plan name	Test points	Run %	Passed %	Failed %	Not run count
Music Playlist Batch Creator - Test Plan	14	100	100	0	0
> TS01 - User Login	4	100	100	0	0
> TS02 - View Playlists	2	100	100	0	0
> TS03 - Real-Time Metadata	2	100	100	0	0
> TS04 - Playlist Editing	4	100	100	0	0
> TS05 - Smart Playlist Creation	2	100	100	0	0



11.Changing the test template

Microsoft Azure Portal | Microsoft Azure DevOps - Microsoft Azure | My Information | Test Plan 84 Music Playlist | Settings - Process (231801095) | Settings - Overview (231801095) |

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings

231801095

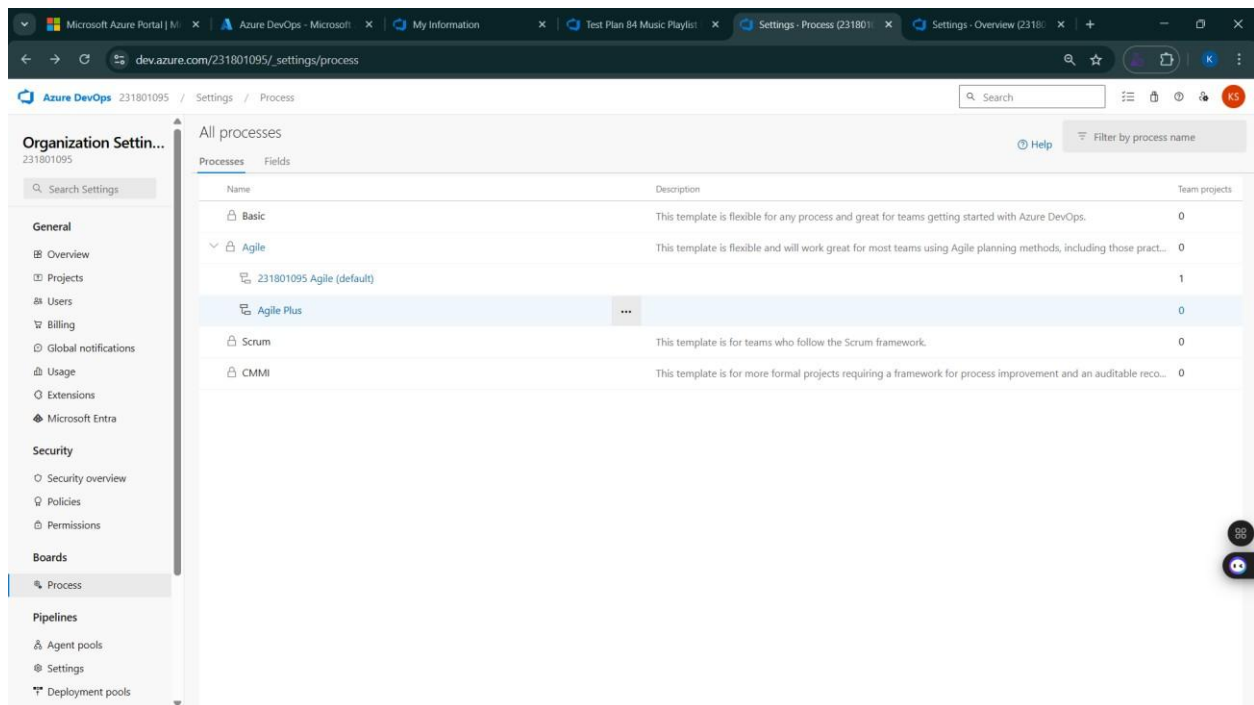
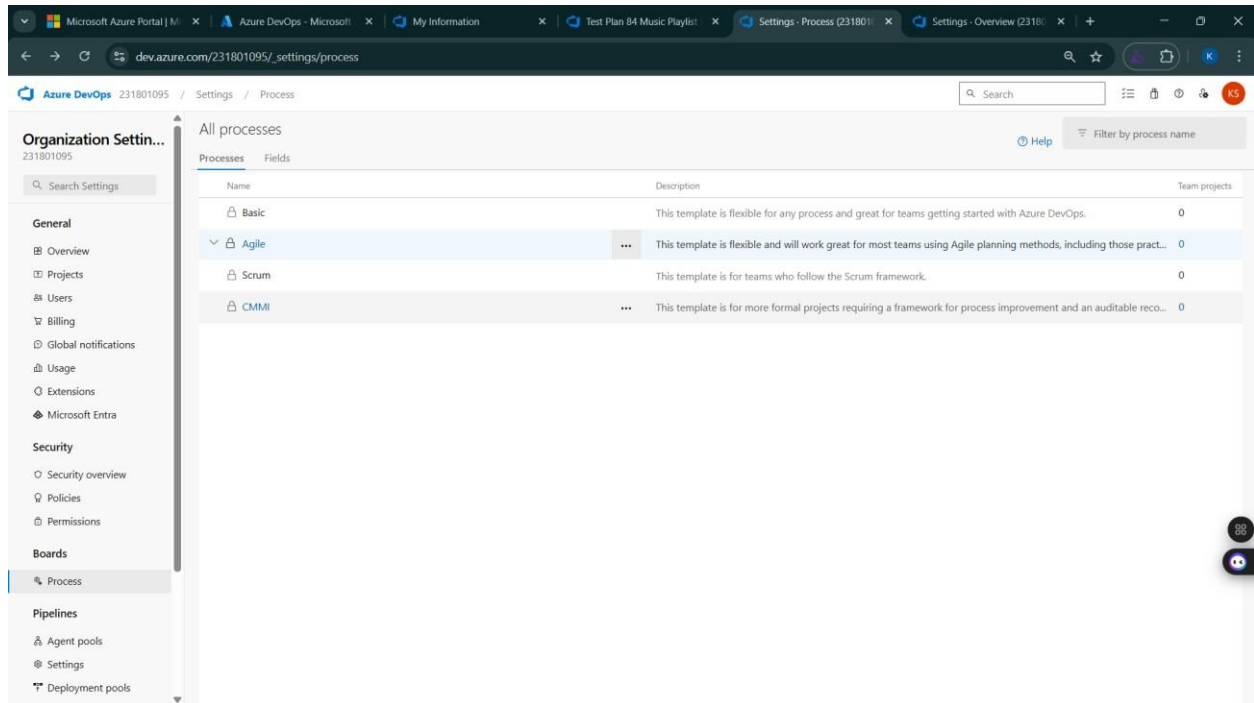
Search Settings

- General
 - Overview
 - Projects
 - Users
 - Billing
 - Global notifications
 - Usage
 - Extensions
 - Microsoft Entra
- Security
 - Security overview
 - Policies
 - Permissions
- Boards
 - Process
- Pipelines
 - Agent pools
 - Settings
 - Deployment pools

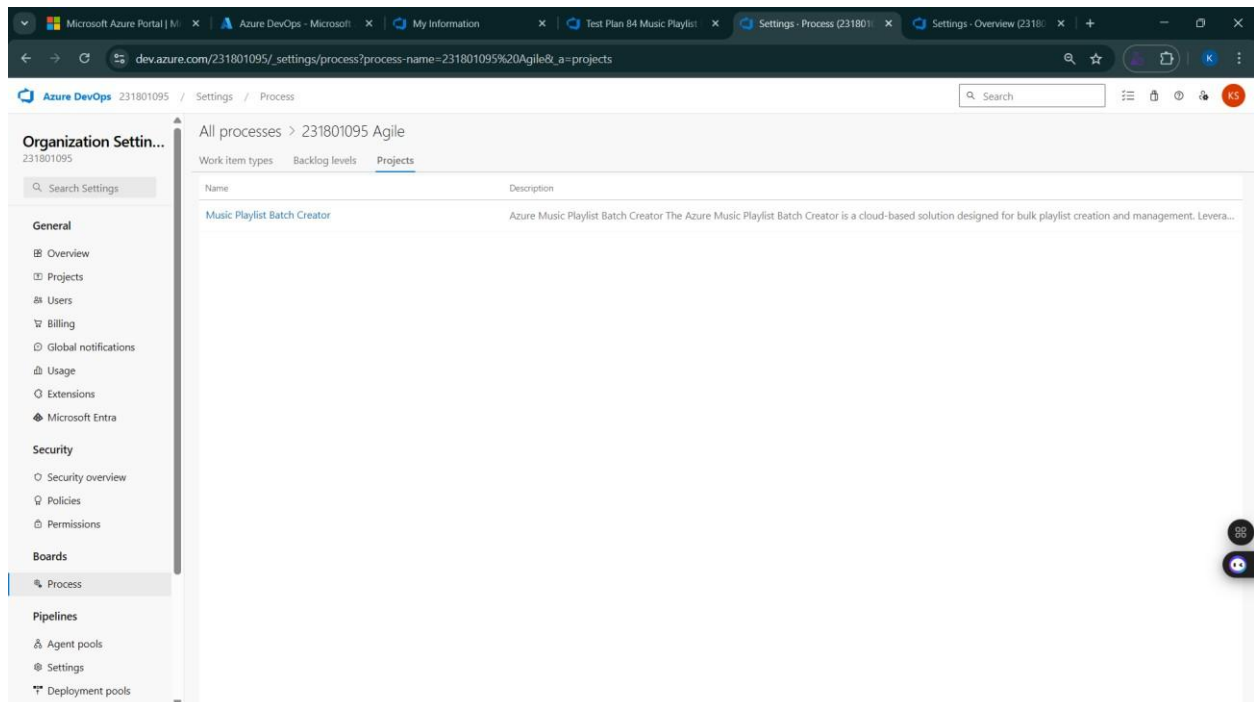
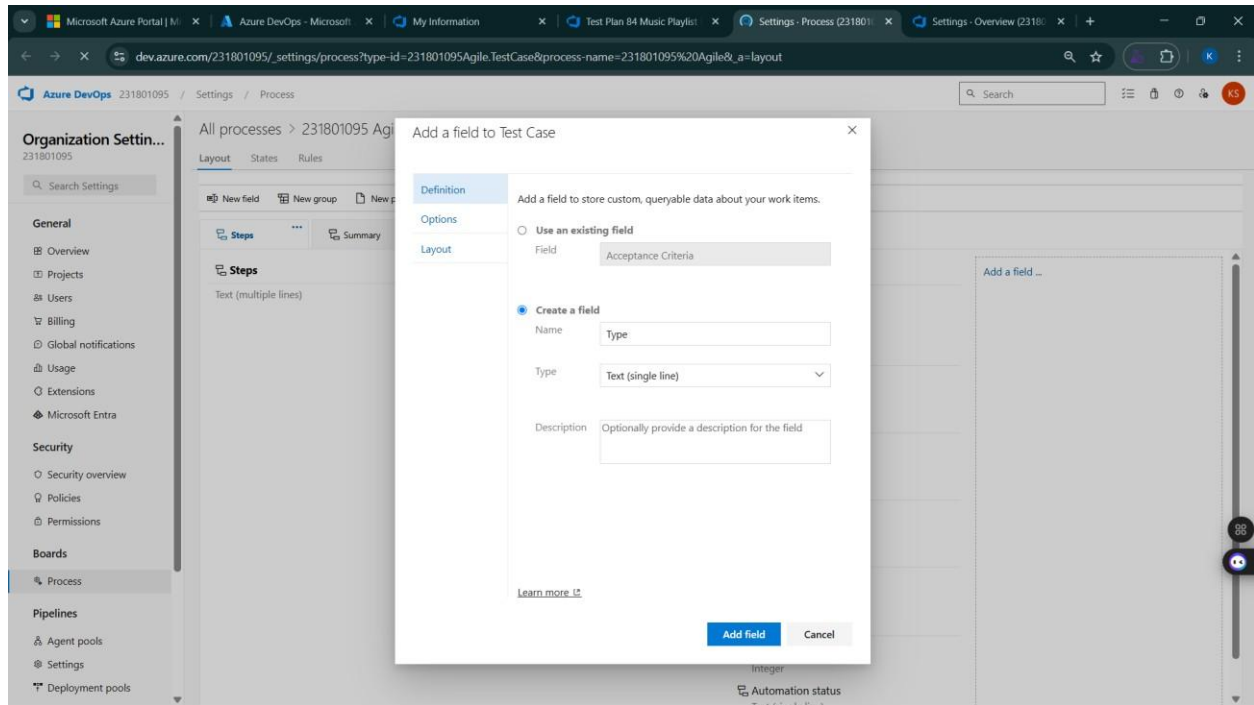
All processes

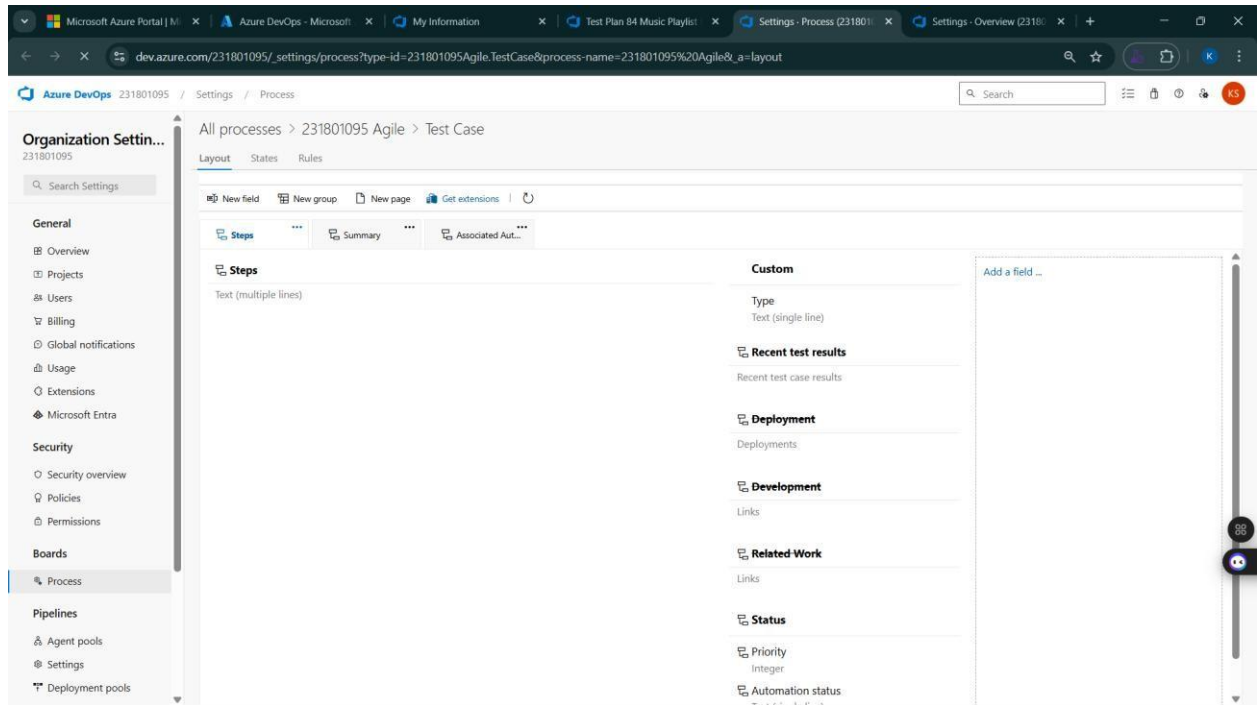
Filter by process name

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0



12.View the new test case template





Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

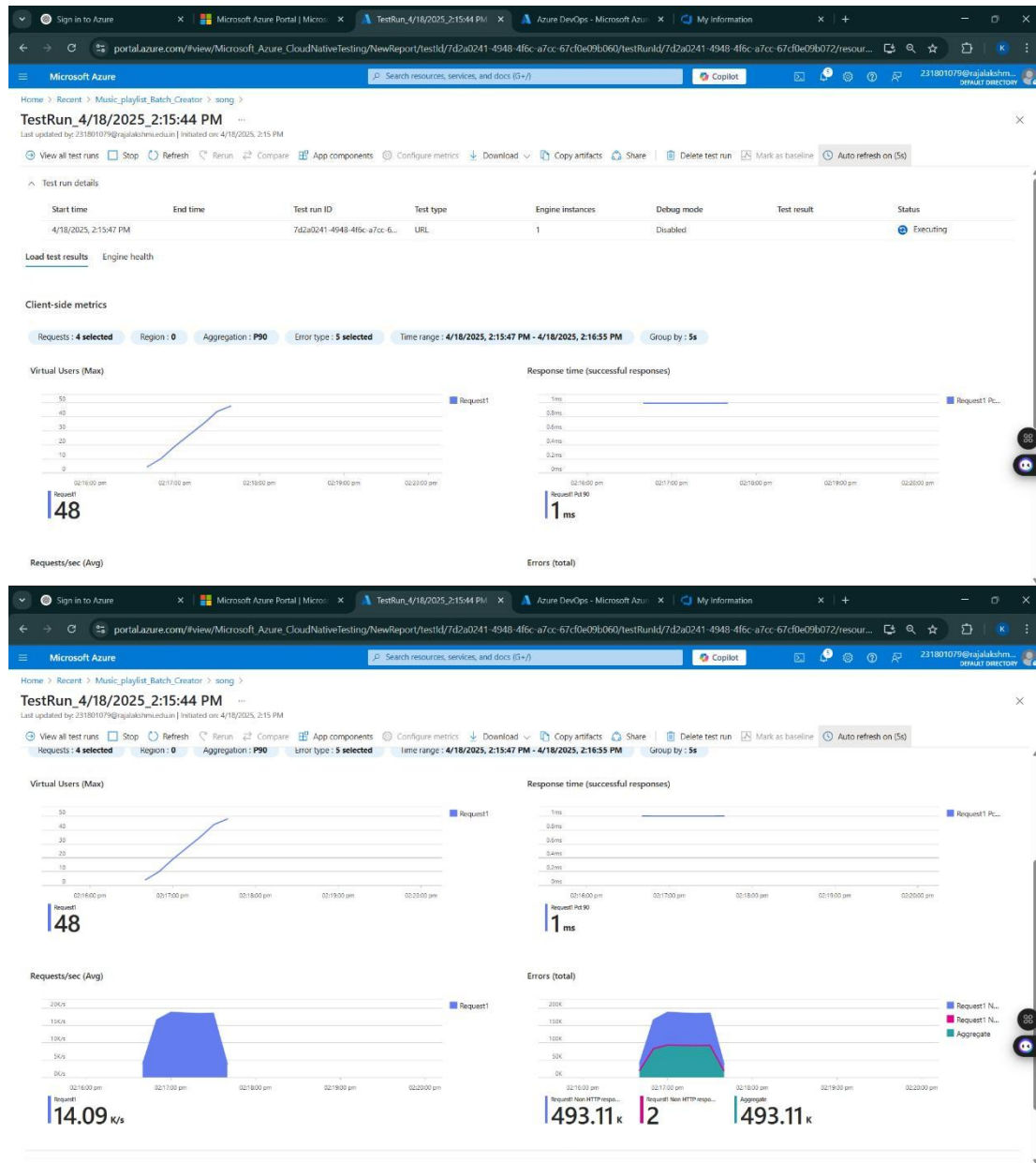
1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource ○ Go to *Create a resource* → Search for “Azure Load Testing”.
 - Select Azure Load Testing and click Create.
3. Fill in the Configuration Details ○ *Subscription*: Choose your Azure subscription. ○ *Resource Group*: Create new or select an existing one. ○ *Name*: Provide a unique name (no special characters).
 - *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

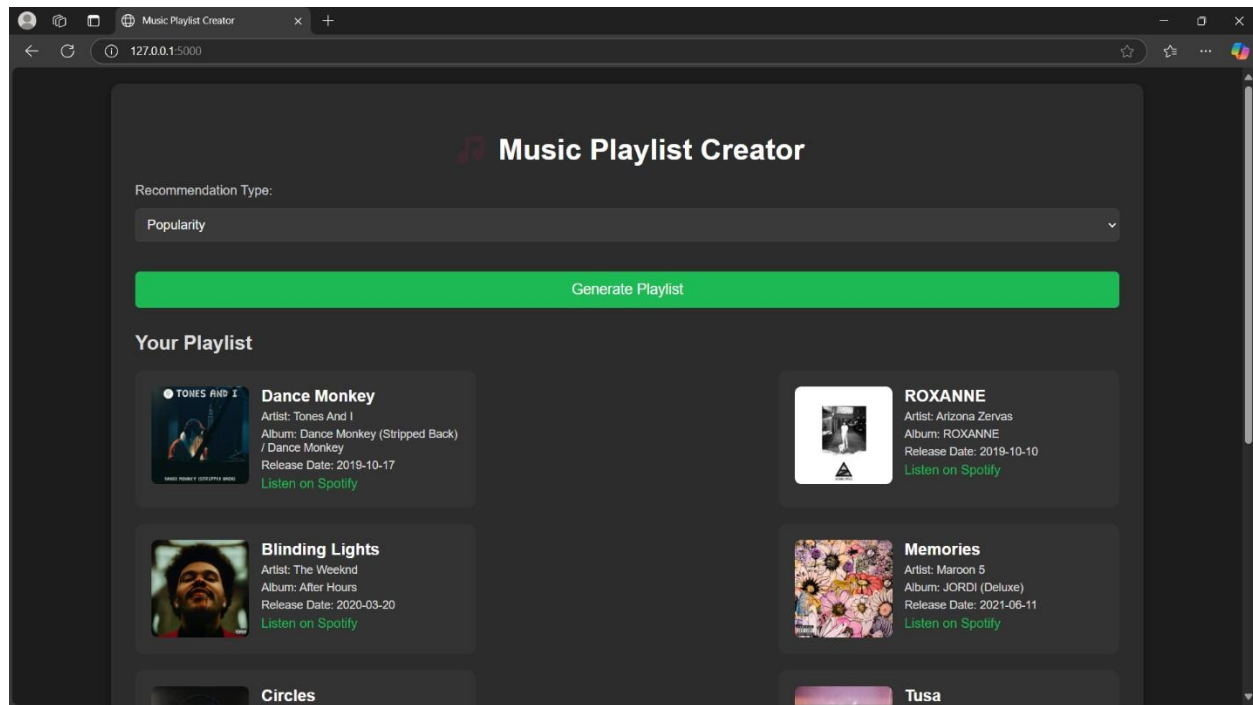
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab ○ *Test Name*: Provide a unique name.
 - *Description*: (Optional) Add test purpose.
 - *Run After Creation*: Keep checked.
3. Load Settings ○ *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - In Azure DevOps, create a new project.
 - Create a pipeline and select GitHub as the source.
 - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - Add the following basic pipeline configuration for Python and Flask:

yml Code trigger:

- main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

Step 1: Checkout the code from GitHub

- checkout: self

Step 2: Set up Python environment

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x' # Use the latest Python 3.x version

displayName: "Set up Python"

Step 3: Install dependencies from the correct path

- script: | python -m pip install --upgrade pip pip install -r project/requirements.txt # Adjusted path to requirements.txt
displayName: "Install dependencies"

Step 4: Run a simple Python script to check the environment

- script: |

python -c "print(' Hello from Music Playlist Batch Creator!')" displayName:

"Run a Python script"

3. Pipeline Tasks Include:

- Setting up the Python environment using the UsePythonVersion task.
- Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

Azure DevOps 231801095 / Music Playlist Batch Creator / Pipelines / Music Playlist Batch Creator... / 20250424.3

#20250424.3 • Pipeline 2
Music Playlist Batch Creator (9)

This run is being retained as one of 3 recent runs by main (Branch). [View retention leases](#)

Summary Code Coverage

Manually run by Karthick S

Repository and version
Music Playlist Batch Creator
main a87bd670

Time started and elapsed
Just now
24s

Related
0 work items
0 artifacts

Tests and coverage
[Get started](#)

[View 52 changes](#)

Jobs

Name	Status	Duration
Job	Success	6s

Project settings

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

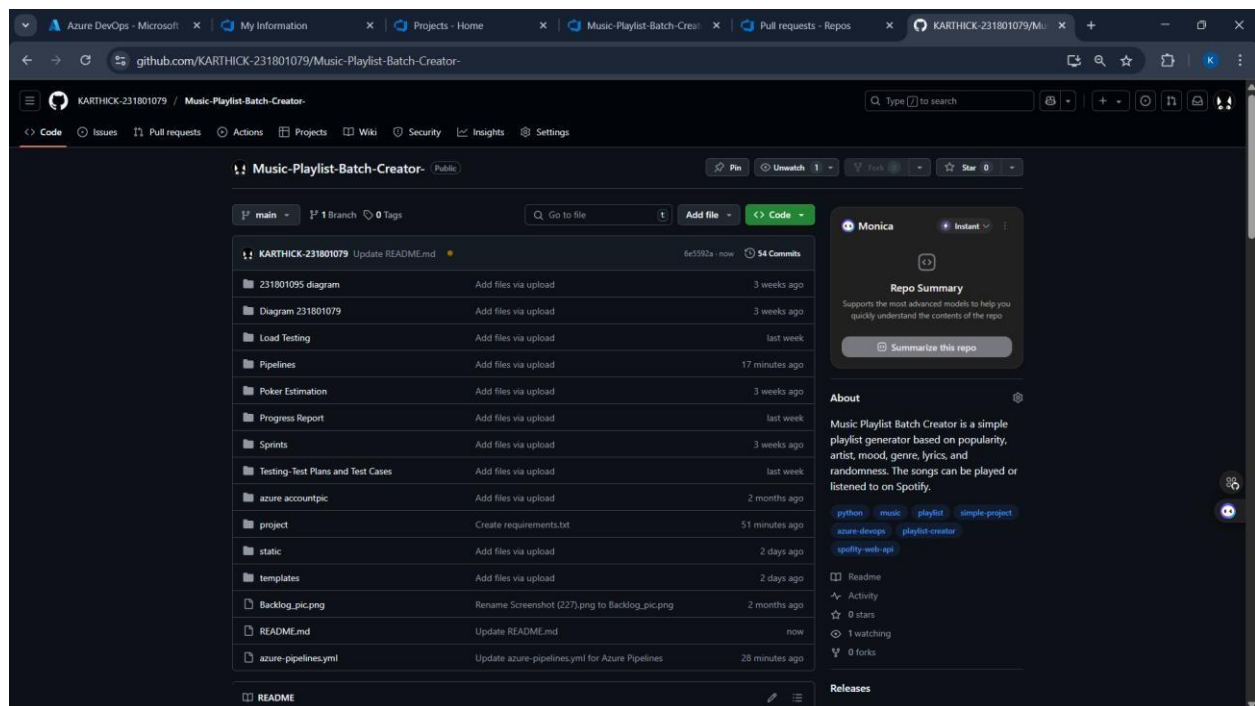
EXP NO: 10

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Music Playlist Batch Creator project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.